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Counterclockwise rotation of Northeast Japan: Paleomagnetic evidence for regional extent and timing of rotation

Yo-ichiro Otofuji [a], Ayumi Kambara [a], Takaaki Matsuda [b], Susumu Nohda [c]

[a] *Department of Earth and Planetary Sciences, Faculty of Science, Kobe University, Kobe 657, Japan*
[b] *Department of Geology, Faculty of Science, Himeji Institute of Technology, Himeji 671-22, Japan*
[c] *Kyoto Sangyo University, Kamigamo, Kita-ku, Kyoto 603, Japan*

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Abstract

Early to Late Miocene welded tuffs have been sampled from northeast Honshu Island and the Oshima Peninsula of Hokkaido Island, Japan for paleomagnetic study. Characteristic directions with a high unblocking temperature component above 560°C are isolated from 24 sites. Westerly declinations ($D = -1° \sim -91°$) are identified in welded tuffs older than 17 Ma from these areas. When combined with previous paleomagnetic data from welded tuffs, the Early Miocene paleomagnetic poles derived from northeast Honshu Island and the Oshima Peninsula are statistically identical and indicate that northeast Honshu Island and the Oshima Peninsula were subjected to counterclockwise rotation as a single tectonic block. These results establish that the rotating block of Northeast Japan extends north to the Oshima Peninsula to the latitude 43°N. The dimension of this rotating block is of the order of 500 km. The paleomagnetic data provide a best curve for the rotation process, with the counterclockwise rotation reaching a climax at 15.0 Ma and a net amount of rotation of 46.4°. The 500 km long Northeast Japan block was rotated counterclockwise through more than 45° and accompanied a clockwise rotation of the 600 km long Southwest Japan block at about 15 Ma.

1. Introduction

Peripheral parts of the continental crust typically behave as a non-rigid assemblage associated with the formation of back-arc basins. As a consequence continental material is often split into blocks which separate from the mother continent [1]. A typical example is observed in the Japan arc and Sea of Japan system (Fig. 1). From the evidence of recent paleomagnetic studies along the Japan arc and ODP results from the Japan Sea [2–6], continental blocks forming the Japan arc are interpreted to have drifted and rotated as a consequence of the opening of the Sea of Japan between 25 and 12 Ma. A number of models have been proposed for the rifting process on the basis of geological and geophysical data derived from the Sea of Japan area during the last decade [e.g., 1,7]. Most models propose that the Japan arc incorporates at least two discrete continental blocks, comprising Southwest (SW) Japan and

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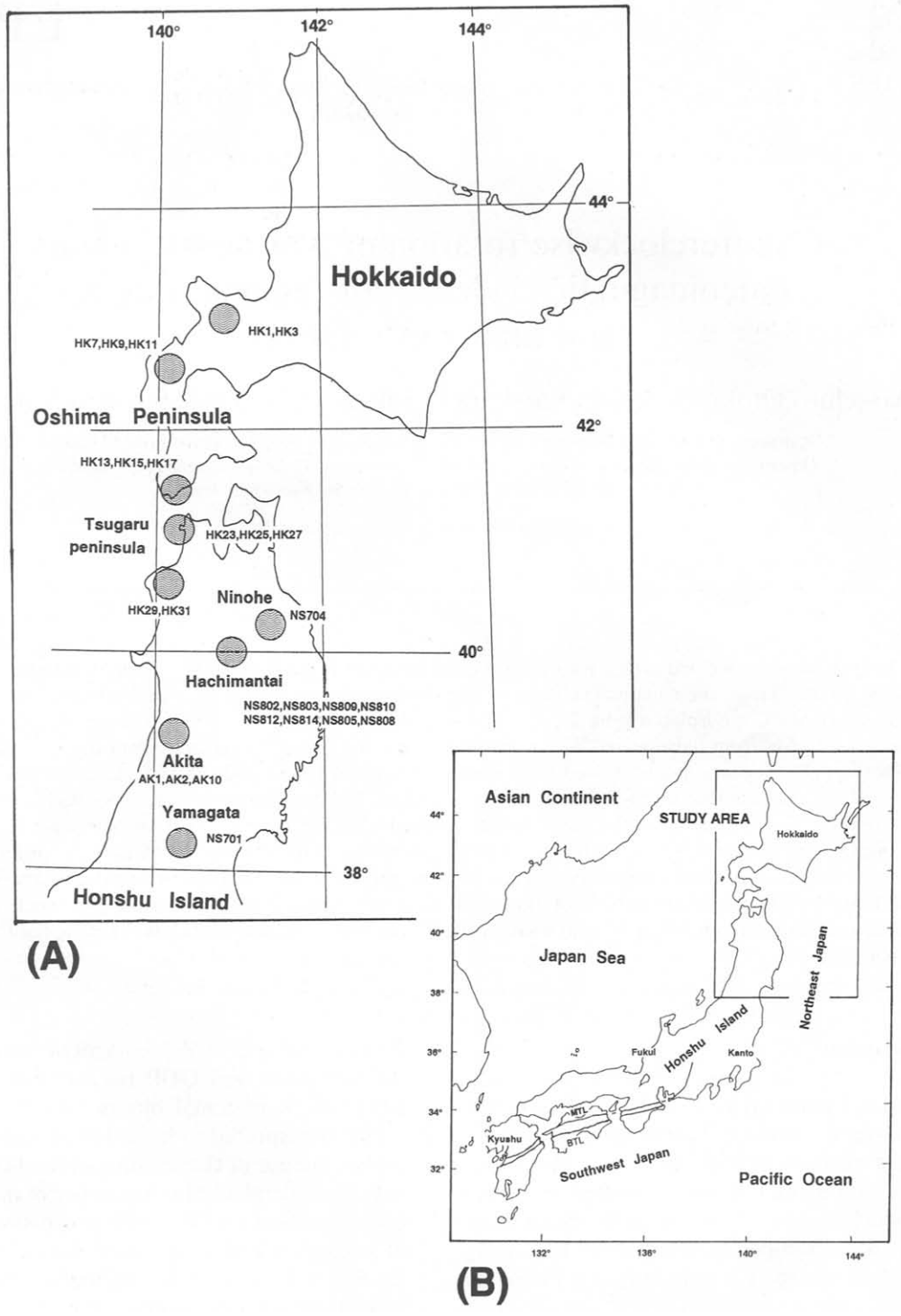


Fig. 1. (A) Map of study area of NE Japan showing location of the paleomagnetic sampling sites. (B) Sketch map of study area, and configuration of the Japan arc and Sea of Japan system. *MTL* = Median Tectonic Line, *BTL* = Butsuzo Tectonic Line.

Northeast (NE) Japan. These two blocks are considered to have experienced differential rotation when the Sea of Japan was opened.

The regional extent of the rigid block comprising SW Japan during this clockwise rotational motion has been clearly established. Itoh and Ito [8] show from paleomagnetic evidence that the area from Kyushu to Fukui behaved as an intact block during the rotation. The length of the rigid block of SW Japan is about 600 km. This intact aspect of this 600 km long continental crust segment is supported by the linear nature of geological structures such as the Median Tectonic Line (MTL) and the Butsuzo Tectonic Line (BTL) (see Fig. 1). In contrast, the extent of the counterclockwise rotation of NE Japan is controversial. While many reconstructions based on geological observations suggest that the rigid area of NE Japan ranges from the Kanto district to the western part of Hokkaido Island (see Fig. 1 and [1,9–12]), most paleomagnetic studies have suggested that only northeast Honshu Island was rotated counterclockwise as an intact block [2,13–15]. Kawai et al. [15] note the possibility of relative rotation between the western part of Hokkaido Island and northeast Honshu Island. These controversies relate to the lack of paleomagnetic data from the northern part of Honshu Island and the western part of Hokkaido Island.

As a step toward resolving this issue we have collected samples from the Yamagata area on Honshu Island to the western part of Hokkaido Island (Fig. 1). The primary subject of our paleomagnetic study is the Oligocene to Late Miocene welded tuffs. These are highlighted because the rotational motion of NE Japan occurred during this period [2,14,16] and because welded tuffs have been shown to provide reliable magnetizations and radiometric ages.

2. Geological setting and sampling

Since Oligocene times NE Japan has been one of the most vigorous Cenozoic volcanic fields seen along convergent plate boundaries [17,18]. The Neogene sedimentary and volcanic rocks overlie a pre-Cenozoic basement which is exposed sporadically in the mountainous areas in northeast Honshu Island and in the western part of Hokkaido Island.

Four different stages can be distinguished in the Cenozoic volcanic activity of NE Japan since Oligocene times [e.g., 6,19]: [(1) Oligocene, (2) Early Miocene, (3) Middle Miocene and (4) Pliocene–Quaternary volcanism].

(1) The Oligocene volcanic (31–28 Ma) is characterized by mafic to felsic volcanic rocks which are distributed through the Sea of Japan side of Honshu Island and in the southwestern part of Hokkaido Island.

(2) At about 22 Ma, intens volcanism began after cessation of the earliest Miocene volcanic activity, and produced a wide range of voluminous volcanic rocks in the same areas in which the Oligocene volcanism was active. Mafic to intermediate volcanic rocks were also produced around 22 Ma, sporadically in the area on the trench side.

(3) While mafic to intermediate volcanic activity continued on the trench side, typical bimodal volcanism followed in early Middle Miocene times on the Sea of Japan side.

(4) Volcanism during Late Miocene to Pliocene times was less intense, although vigorous volcanism was reactivated from latest Pliocene times to the Quaternary throughout NE Japan.

We focus our paleomagnetic study on the welded tuffs of these four stages of volcanism. More than 300 samples were collected from 26 sites at 9 localities in 6 areas that have been mapped in detail: (1) the Oshima Peninsula of Hokkaido Island [20–25], (2) the Tsugaru Peninsula [23,26], (3) Hachimantai [27], (4) Ninohe [28], (5) Akita [23,29] and (6) Yamagata [23,30]. The sampling localities are shown in Fig. 1 and listed in Table 1.

Eutaxitic structure (lineation of stretched pumice and aligned phenocrysts) was identified and recovered at each outcrop of the welded tuffs. Post-emplacement tilting of the welded tuffs was adjusted for on the basis of this. However, the eutaxitic structure does not always indicate the paleohorizontal because this may have been formed parallel to a primary slope, and it is observed that modest errors are always present in

Table 1
Summary of paleomagnetic results for the Oligo-Miocene welded tuffs from the Oshima Peninsula and northeast Honshu Island of NE Japan

The magnetization directions of the high-temperature component are listed. Demagnetization: A = single-component magnetization with high-temperature component; B = distinctive two-component magnetizations; C = no high-temperature component was isolated. Temperature range used to calculate the characteristic direction by principal component analysis is shown in the column headed "°C". N = number of samples oriented in the field. D and I = declination and inclination respectively; k = Fisherian precision parameter; α_{95} = radius of cone of 95% confidence within site. Mean ages are calculated by assigning unit weight to an age datum found by the K-Ar and fission-track method. Rock type: D = dacite welded tuff; DT = dacite tuff; PA = pyroxene andesite welded tuff; R = rhyolite welded tuff. Data denoted by asterisk are from Otofuji et al. [2].

High temperature component

[Area] Formation site	(age) Demagnetization (°C)	N	in-situ D (°)	in-situ I (°)	tilt-corrected D (°)	tilt-corrected I (°)	k	α_{95} (°)	Rock Type	Locality Lat. (E)	Locality Long. (N)
[Oshima Peninsula]											
Kunitomi Formation (24.7± 1.2 Ma [54])											
HK1	B 0-600	8	-72.9	72.6	-63.1	53.1	293.8	3.2	D	140°52.5'	42°59.1'
HK3	B 100-630	11	-88.5	77.5	-67.1	76.5	102.5	4.5	D	140°52.6'	42°59.0'
Masukawa Formation =mean 16.9 Ma (15.1, 17.2 Ma [25]; 18.3±1.1Ma [54])											
HK7	B 150-600	11	141.5	-49.5	135.4	-73.3	197.3	3.3	R	140°06.7'	42°36.0'
HK9	B 150-600	11	155.4	-45.2	178.6	-46.9	154.9	3.7	R	140°06.8'	42°35.4'
HK11	B 150-600	13	151.1	-41.2	167.3	-50.3	51.3	5.9	R	140°06.8'	42°35.5'
Fukuyama Formation =mean 23.0 Ma (21.7, 22.3, 24.3 Ma [22]; 24.6, 23.3, 20.8 Ma [21]; 24.2, 23.1 Ma [55]; 24 Ma[20])											
HK13	B 200-600	6	86.1	-2.7	93.5	-42.5	165.4	5.2	R	140°13.5'	41°25.1'
HK15	B 200-630	13	86.7	9.6	92.2	-43.2	21.1	9.2	R	140°13.0'	41°25.5'
[Tsugaru Peninsula]											
Gongenzaki Formation=Early Miocene											
HK23	A 100-630	12	106.5	-24.8	104.1	-45.5	182.4	3.2	D	140°16.7'	41°08.1'
HK25	B 150-630	12	108.2	-16.5	112.0	-52.2	52.6	6.0	D	140°16.6'	41°08.1'
HK27	A 150-630	6	108.0	-29.0	140.3	-66.8	367.2	3.5	D	140°16.8'	41°08.1'
Ohtose Formation (16.8 Ma [26])											
HK29	B 200-600	11	175.4	80.6	-51.9	67.3	233.0	3.0	D	140°08.2'	40°38.7'
[Ninohe]											
Nisatai Formation (21.8 Ma, [56])											
NS704	B 300-590	12	-90.1	29.7	-91.3	44.6	126.2	3.4	D	141°20.8'	40°20.8'
NS630*	500	10	-77.3	52.7	-67.2	39.8	446.9	2.3	D	141°20.0'	40°17.3'
[Hachimantai]											
Itsukaichi Formation=mean 6.3 Ma (6.1, 6.7, 5.6,6.9 Ma [27])											
NS802	A 0-600	12	176.4	-46.6	176.4	-46.6	256.8	2.7	D	141°06.9'	40°01.7'
NS803	A 0-630	12	-177.9	-52.8	169.9	-54.0	266.6	2.7	D	141°06.1'	40°00.8'
Azukizawa Formation=mean 7.15 Ma (7.0, 7.3 Ma [27])											
NS809	A 0-620	9	13.9	54.1	-4.7	66.5	302.7	3.0	D	140°51.5'	40°07.7'
NS810	A 0-600	11	8.0	53.5	-3.2	62.1	207.1	3.2	D	140°51.4'	40°07.6'
NS812	A 0-600	12	11.1	57.8	22.3	45.5	123.0	3.9	D	140°51.4'	40°07.1'
NS814	A 0-620	11	9.0	62.8	10.6	75.6	757.0	1.7	D	140°51.7'	40°07.1'

Table 1 (continued)

Kiritoshi Formation= mean 11.6 Ma (12.8, 10.4 Ma [27])												
NS805	A	0-600	12	-167.2	-41.6	-167.2	-41.6	315.8	2.5	D	140°56.1'	40°00.8'
NS808	A	150-600	12	-179.4	-45.7	-179.4	-45.7	56.6	5.8	D	140°56.4'	40°00.8'
[Oga Peninsula]												
Daijima Formation = 20.1 ± 1.1 Ma (18.8 Ma [56]; 21.5 Ma [57]; 21.5 Ma [58])												
NS311*		400	6	-54.7	45.5	-29.1	57.1	643.7	2.6	D	139°45.8'	39°51.4'
NS633*		560	11	-54.2	42.2	-31.6	40.9	163.3	3.6	D	139°45.8'	39°51.3'
[Taiheizan]												
Okuramata Formation (14.1 Ma [2])												
NS413*		400	8	-15.2	24.1	-10.6	42.9	140.9	4.7	R	140°23.3'	39°43.1'
NS417*		400	10	-10.8	46.1	-9.9	61.1	92.2	5.1	R	140°23.8'	39°42.5'
[Akita]												
Yashiozawagawa Formation (17.2 ± 0.9 Ma [54])												
AK1	B	200-630	12	108.0	-25.5	116.8	-53.9	102.5	4.3	R	140°12.6'	39°13.9'
AK2	B	150-630	12	127.0	-46.6	138.3	-54.3	527.0	1.9	R	140°12.3'	39°13.5'
AK10	B	150-630	10	109.9	-36.5	125.0	-67.9	196.3	3.5	R	140°12.4'	39°14.0'
[Yuzawa]												
Zentoku Welded Tuff (21.9 Ma [56])												
NS512*		300	10	150.1	-29.3	132.7	-33.8	178.9	3.6	R	140°19.1'	39°15.7'
Masikozawa Welded Tuff (21.9 Ma [56])												
NS513*		300	11	157.1	-70.1	134.9	-73.6	174.6	3.5	PA	140°19.9'	39°11.0'
NS515*		300	6	166.3	-69.3	146.5	-74.1	77.4	7.7	PA	140°20.0'	39°10.9'
NS516*		300	8	136.1	-66.2	140.0	-76.1	239.7	3.6	PA	140°19.6'	39°11.2'
Yunosawagawa Formation (33.0 Ma [56])												
NS426*		300	9	-36.1	79.8	-56.1	67.9	361.2	2.7	PA	140°22.1'	39°06.7'
NS427*		300	9	-45.4	68.4	-41.8	48.5	713.4	1.9	PA	140°21.9'	39°07.0'
NS518*		300	10	-61.6	67.2	-31.8	72.9	330.3	2.7	PA	140°24.6'	39°01.4'
Okatsugawa Formation (25Ma [59]) Minamizawa Welded Tuff												
NS430*		300	10	138.2	-46.6	127.4	-48.4	323.3	2.7	D	140°22.1'	39°02.1'
NS520*		300	8	162.6	-58.2	178.2	-75.1	26.9	10.9	D	140°21.7'	39°01.6'
NS522(A)*		300	6	141.4	-38.9	133.2	-41.4	82.9	7.4	D	140°21.9'	39°01.9'
NS522(B)*		300	9	156.7	-47.1	138.6	-50.8	149.8	4.2	D	140°21.9'	39°01.9'
NS525*		500	8	165.2	-57.0	143.1	-54.7	48.1	8.1	D	140°21.3'	39°00.9'
NS527*		300	12	175.2	-49.3	142.2	-53.1	450.4	2.0	D	140°21.6'	39°00.3'
[Northern Asahi]												
Higashi-Ohotorigawa Formation (21 Ma [2])												
NS317*	♀	400	12	172.0	-16.7	152.5	-50.5	322.1	2.4	PA	139°46.5'	38°28.1'
[Yamagata]												
Narasawa Formation (11.3 Ma [59])												
NS701	B	0-620	12	-178.1	-63.2	167.0	-61.3	750.1	1.6	D	140°25.8'	38°19.1'
NS609*		400	12	-171.3	-54.1	165.5	-66.9	175.4	3.3	D	140°25.2'	38°20.7'
NS611*		400	11	-167.2	-66.7	166.5	-71.8	608.0	1.9	D	140°25.4'	38°20.3'

measurements of dip (< 5°) and strike (±5°). Thus, sampling in each welded tuff unit was carried out at widely spaced outcrops to reduce the error caused by this uncertainty in tectonic correction.

The welded tuffs were collected as hand samples oriented by magnetic compass. Each paleomagnetic site typically comprised twelve independently oriented samples, distributed over distances ranging up to 15 m.

3. Paleomagnetism

In the laboratory, individual specimens 25 mm in diameter and 23 mm long were prepared from the samples. Natural remanent magnetizations (NRMs) were measured with either a spinner magnetometer or an ScT cryogenic magnetometer, depending on the intensity of the magnetization. All specimens were subjected to stepwise thermal demagnetization up to 630°C. Results for each specimen were plotted on orthogonal vector diagrams [31] to assess component structure and on equal-area projections to evaluate directional stability. Principal component analysis [32] was used to estimate the component directions.

Progressive thermal demagnetization of the samples separates the 26 sampling sites into three categories (Fig. 2, Fig. 3, Table 1 and Table A1).

(Category A): Ten sites exhibit a single mag-

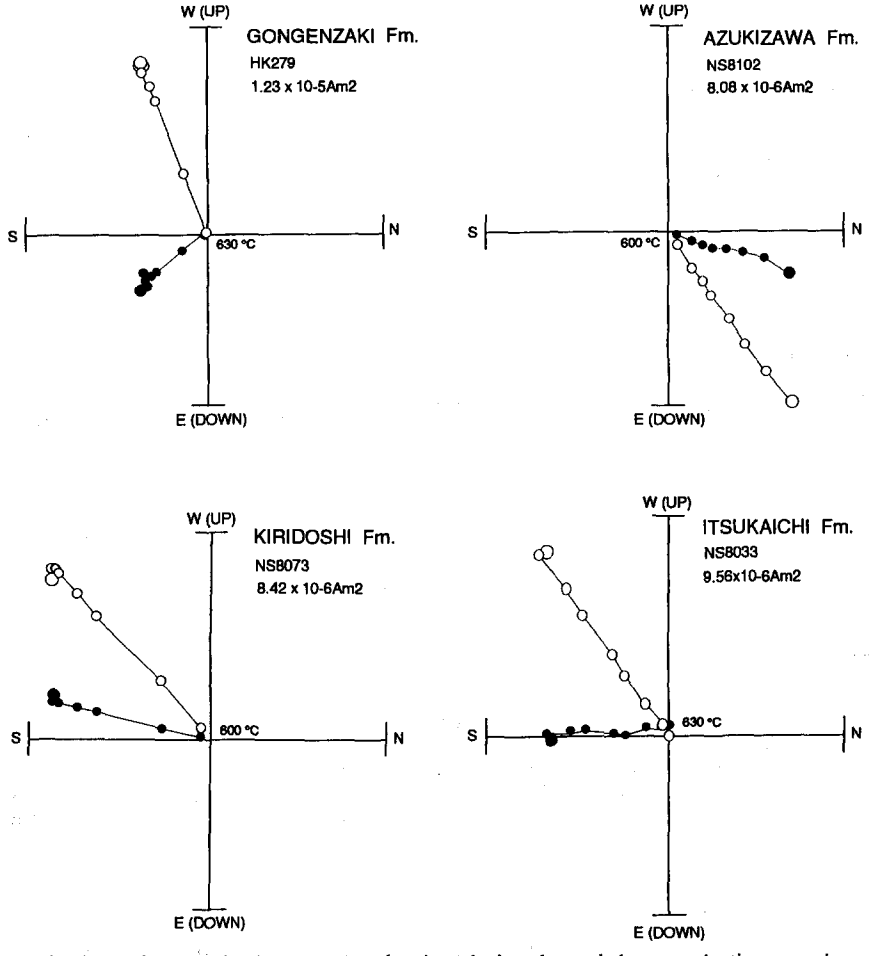


Fig. 2. Orthogonal projections of magnetization vector end points during thermal demagnetization experiments for specimens in category A (see text). Open symbols show the magnetization vector projection on the vector plane, solid symbols on the horizontal plane. Tectonic tilt correction has been applied.

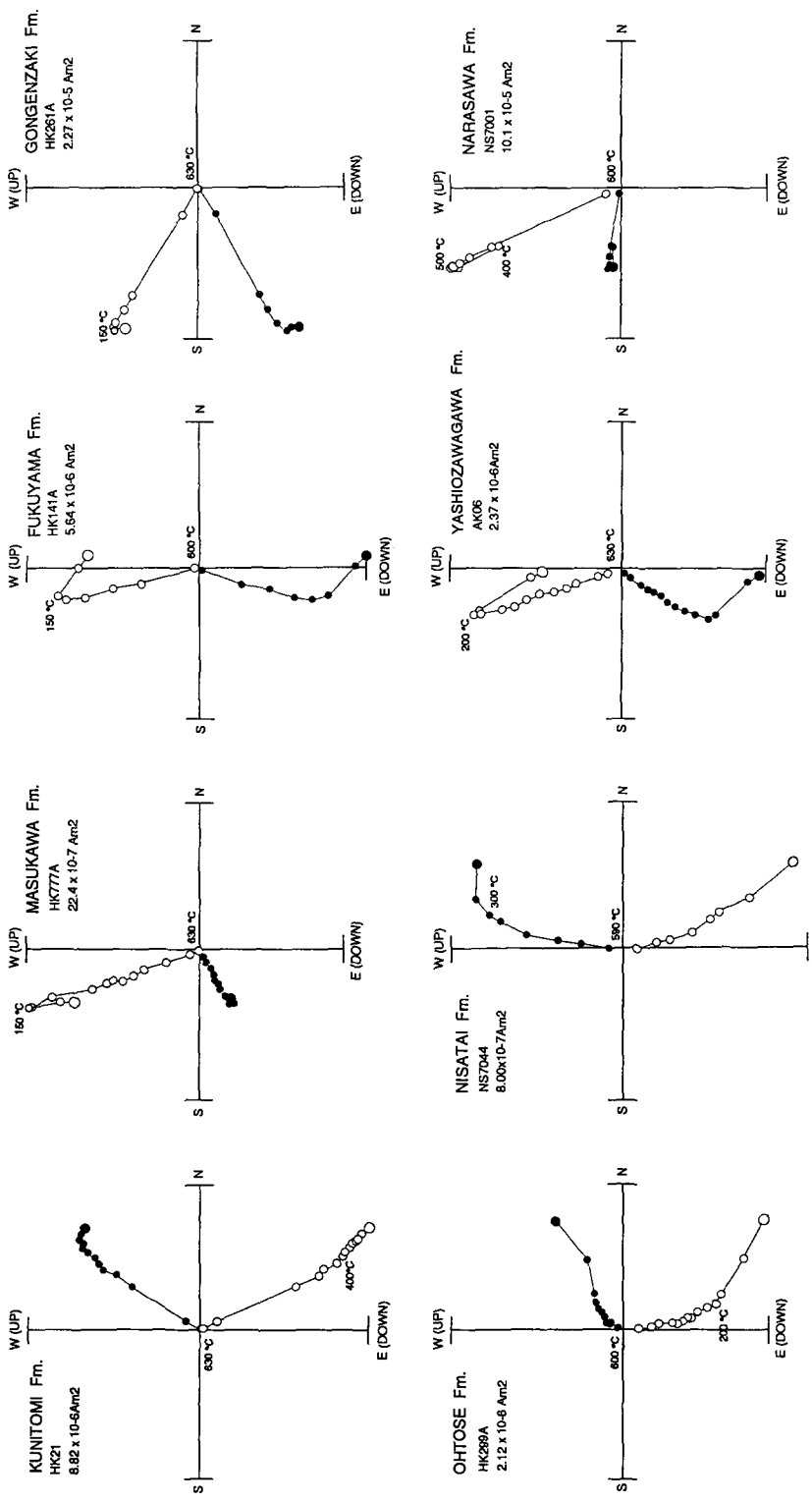


Fig. 3. Orthogonal projections of magnetization vector end points during thermal demagnetization experiments for specimens in category B (see text). Open symbols show the magnetization vector projection on the vector plane, solid symbols on the horizontal plane. Tectonic tilt correction has been applied.

netic component with unblocking temperatures of 600–630°C (Fig. 2). Little change is observed in the direction of remanent magnetization after demagnetization at temperatures higher than the Curie temperature of magnetite. Both magnetite and hematite probably carry the stable remanent magnetization with almost the same direction.

(Category B): Fourteen sites reveal two distinct magnetization components and a high unblocking temperature component is isolated (Fig. 3). The high-temperature component ranges between 590° and 630°C, indicating both magnetite and hematite carriers. The low-temperature components are removed from twelve sites in the initial stage of demagnetization by 400°C (Table A1). Their *in-situ* directions are dominated by a component in the present direction of the Earth's magnetic field which is therefore probably a vis-

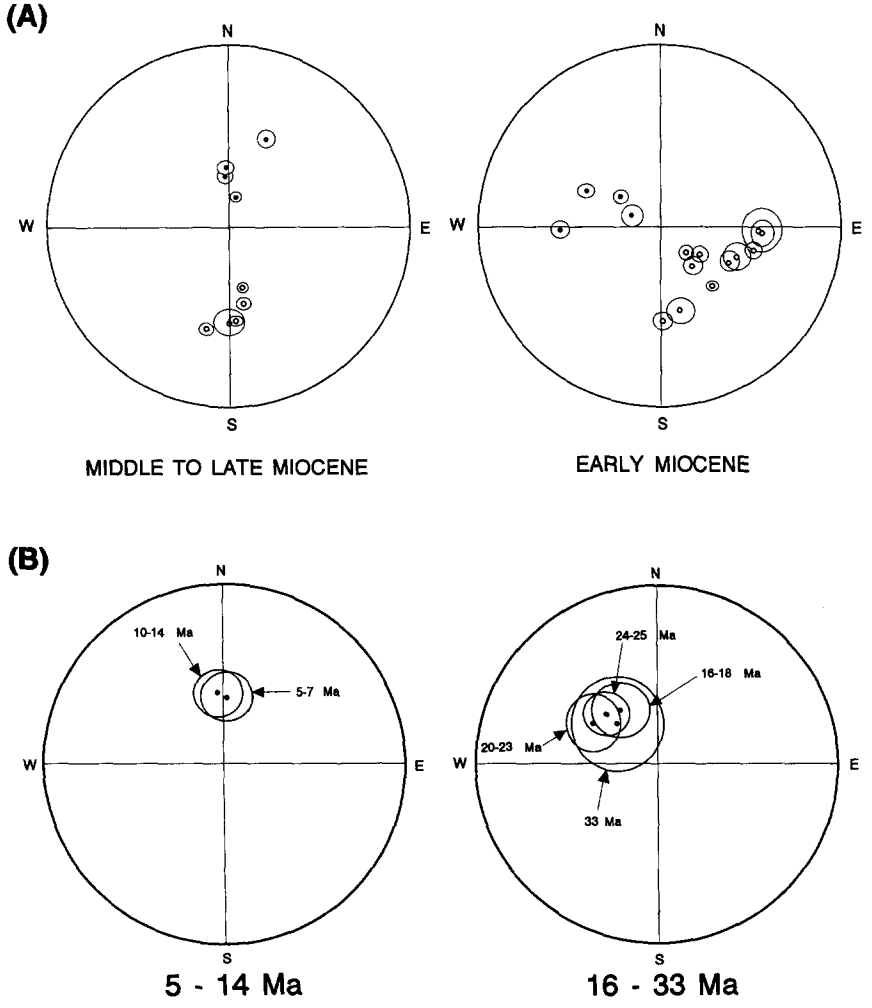


Fig. 4. (A) Paleomagnetic directions for NE Japan from the present study. Equal-area projections of the site mean direction with a 95% confidence circle after tilt correction for the Miocene welded tuffs from northeast Honshu Island and the Oshima Peninsula of Hokkaido Island. Data are divided into two periods, Early Miocene and Middle–Late Micene. The solid symbols refer to the lower hemisphere, the open symbols refer to the upper hemisphere. (B) Summary of reliable paleomagnetic results of the welded tuffs from NE Japan. The mean paleomagnetic directions with 95% confidence circles are shown on equal-area projection for six time intervals: 5–7 Ma, 10–14 Ma, 16–18 Ma, 20–23 Ma, 24–25 Ma and 33 Ma.

cous partial thermoremanent magnetization acquired in recent times [33].

(Category C): From two sites no high-temperature component was obtained (site 31 and 17). Although site 31 in the Ohtose Formation reveals straightforward demagnetization behavior, its magnetization converges to the origin by about 400°C. The *in-situ* direction of this component is similar to the present axial geomagnetic field (Table A1). As a non-welded aspect is observed at this site, no higher temperature component may have been acquired. This low unblocking temperature component is probably of post-folding origin and due to a thermoviscous remanent magnetization acquired during the Brunhes normal polarity chron [33].

Site 17 of the Fukuyama Formation has two components of overlapping blocking temperature spectra above 400°C. The remanent magnetization direction migrates, but does not reach a stable end point. No high-temperature component was isolated.

4. Paleomagnetic results

4.1. Characteristic paleomagnetic direction

Twenty-four sites out of 26 preserve the high-temperature component of magnetization. The 95% confidence circles of the site mean directions of this component do not exceed 10° for any of the 24 sites (Table 1). The direction of the high-temperature component after tilt correction is recognized as a characteristic paleomagnetic direction for each site.

Geological formations with these high-temperature characteristic directions comprise two age groups, Early Miocene and Middle to Late Miocene (Fig. 4A). The declinations of all Early Miocene sites are characterized by counterclockwise deflection from the north or the south in both normal and reversed polarities. The amount of deflection ranges between 1° and 89°. The inclination ranges from 43° to 76°.

Four Middle to Late Miocene sites show paleomagnetic directions with normal polarity and the remaining five sites are of the reversed polarity. Declinations are directed north and south with inclinations ranging from 41° to 72°.

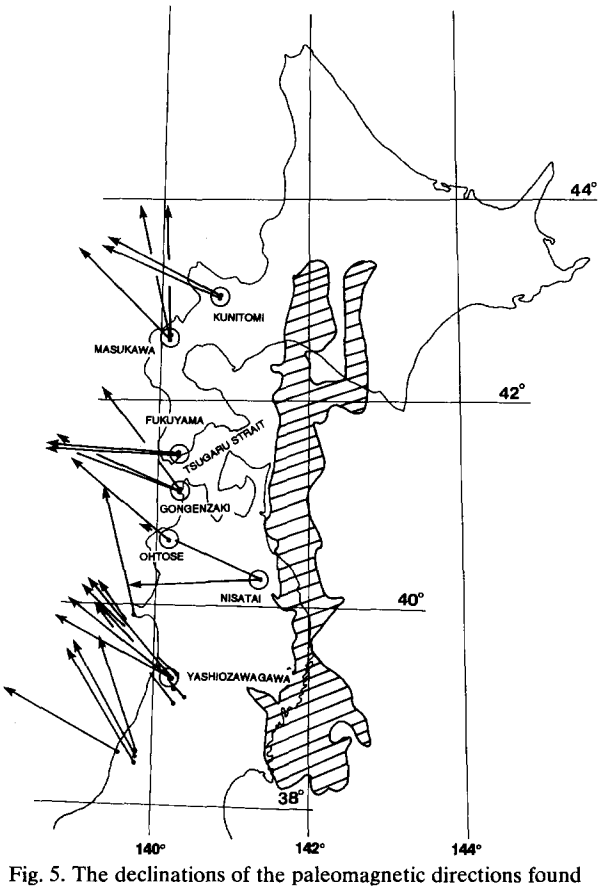


Fig. 5. The declinations of the paleomagnetic directions found for Oligocene and Early Miocene welded tuffs of NE Japan. The data from the present study are shown for the encircled sampling sites; the other data are from the previous study of Otofuji et al. [2]. The positive magnetic anomaly belt (shaded area) runs from the eastern side of northeast Honshu Island to the southwestern part of Hokkaido Island and has a maximum amplitude that exceeds 500 nT [35].

4.2. Geographical distribution of the westerly paleomagnetic direction

The Declinations of Early Miocene formations are plotted in Fig. 5 together with previous results from the Oligocene and Early Miocene welded tuffs [2]. Reversed directions have been inverted. The present results are shown with dots and circles and the results from previous studies

are shown as dots. It is apparent that paleomagnetic data from welded tuffs throughout this sector of NE Japan are characterized by westerly directions. While previous study identified westerly declination values from latitudes 38°21′N–40° 17′N, the present study indicates that the larger area north to 42°59′N is characterized by the counterclockwise deflection in declination of Early Miocene rocks.

The westerly direction is now also recognized on the Tsugaru Peninsula, the westerly declination appearing in the Ohtose Formation (16.8 Ma) (–40°) and in the Gongenzaki Formation of strongly welded tuffs (–76°). Northernmost Honshu Island north of 41°N is therefore characterized by the westerly declination in the Early Miocene rocks.

The counterclockwise deflection in declination is now recognized in the Oshima Peninsula on Hokkaido Island on the north side of the Tsugaru Strait, specifically within the Fukuyama Formation (23.0 Ma) in which paleomagnetic directions are characterized by the largely counterclockwise deflection in declination ($D = -87°$, $-88°$). A westerly direction has previously been reported from welded tuffs in the Fukuyama Formation ($D = -37°$, $-72°$) [34], although little information about the paleomagnetic treatment was described in the study that found these values. On the northern edge of the study area the Kunitomi Formation (24.7 Ma) has declination values of –63° and –67° with normal polarity at two sites.

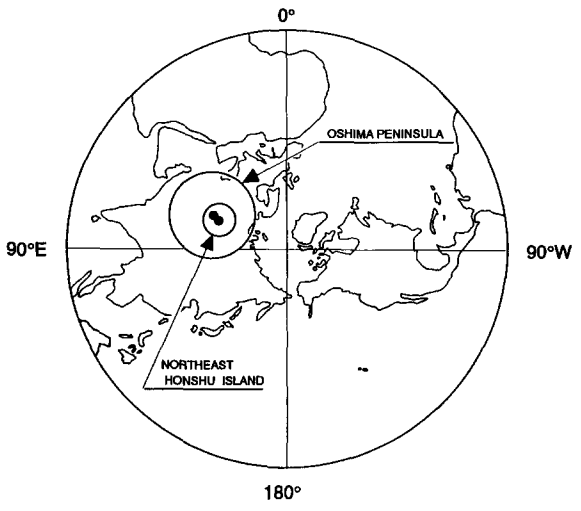


Fig. 6. Early Miocene paleomagnetic poles (with associated 95% confidence circles) for the Oshima Peninsula of Hokkaido Island and northeast Honshu Island. Polar stereographic projection.

5. Discussion

5.1. Extent of the rigid block that underwent counterclockwise rotation

The area from latitudes 38° to 43°N including northeast Honshu Island and the Oshima Peninsula is thus characterized by westerly declinations of remanence in Early Miocene welded tuffs. The directions are identified in normal and reversed polarities by thermal demagnetization. This result is ascribed to the counterclockwise rotation of NE Japan.

Although the Oshima Peninsula and northeast Honshu Island is geographically divided by the Tsugaru Strait, westerly declinations are observed in Early Miocene rocks from both areas. We compare the paleomagnetic directions from the two areas to detect possible relative movement between them. The paleomagnetic poles for Early Miocene rocks from the Oshima Peninsula (41°25′–42°59′N) and northeast Honshu Island (38°28′–41°08′N) are shown in Fig. 6. We have combined two paleomagnetic data from welded tuffs of the Fukuyama Formation reported by Fujiwara et al. [34]. The paleomagnetic pole from the Oshima Peninsula is at 50.4°N, 64.7°E (A_{95} = 19.9°), whereas the paleomagnetic pole from northeast Honshu Island is at 53.3°N, 66.3°E (A_{95} = 7.8°). Although the 95% confidence circle of the pole from Oshima Peninsula is fairly large, the two poles are statistically coincident. Because the Oshima Peninsula and northeast Honshu Island preserve indistinguishable westerly paleomagnetic directions, we interpret this to mean that these areas rotated as a single rigid block in paleomagnetic terms.

Kawai et al. [15] claim that the Oshima Peninsula was rotated clockwise with respect to northeast Honshu Island. This hypothesis was based on the observation that the Neogene paleomagnetic directions of the Oshima Peninsula are characterized by an easterly declination ($D = 18°$), but is contradicted by recognition of westerly directions in Early Miocene rocks. We also identify an easterly direction in the low-temperature components from welded tuffs of the Oshima Peninsula. However, because the *in-situ* direction of this component is parallel to the present field, the easterly direction only appears after tilt correction (Table A1). An easterly declination is expected after tilt correction in insufficiently demagnetized samples from the post-Miocene rocks of the Oshima Peninsula.

The counterclockwise rotating block of NE Japan extends north to 43°N on the Oshima Peninsula. This rotating block is approximately 500 km long, which is compatible with the intact 600 km length of SW Japan. The existence of an extended 500 km long rigid block is inferred from the magnetic anomaly observed in NE Japan [35–37] (see Fig. 5). Segawa and Oshima [35] identify this N–S trending linear positive anomaly belt running from the eastern side of northeast Honshu Island to the southwestern part of Hokkaido Island. The extent of the magnetic anomaly ranges between the latitudes 37.8°N and 43.4°N and can be modelled by a magnetization with a declination of −45° and inclination of 53° [35]. Because the anomaly is coherent the implication is that a block of 600 km in length rotated as a single unit. The extent of the magnetic anomaly is similar to the range in distribution of the westerly paleomagnetic directions of the Early Miocene (Fig. 5).

This conclusion is also supported by the linear distribution of Oligocene to Miocene volcanic rocks in NE Japan [18,38]. Oligocene volcanics (28–31 Ma) are distributed along the west coast (i.e. the Sea of Japan side from Kitaoguni (37.5°N) through the northern areas of Honshu Island and Okujiri Island to the Oshima Peninsula across the Tsugaru Strait. A linear trend is apparent, even though the volumes of these rocks are small. Voluminous Early Miocene volcanic rocks are fairly continuously distributed along the back-arc side of the present NE Japan arc from 37°N to the north, so that the eastern edge of the distribution of volcanic rocks has the shape of a gentle curve. These trends in the distribution of volcanic rocks have been recognized as Oligocene and Early Miocene volcanic fronts [19] and have enabled the identification of the area from northeast Honshu Island to the Oshima Peninsula as having behaved as a single tectonic block since Early Miocene times.

5.2. Timing of counter-clockwise rotation of the Northeast Japan block

The rotation can be recorded as a function of age on the basis of the paleomagnetic data from the dated welded tuffs in the single tectonic block of NE Japan. The K-Ar and zircon fission-track ages for Oligo-Miocene formations of NE Japan are compiled in Table 1.

The paleomagnetic directions are plotted in Fig. 4B. The paleomagnetic directions of the sampling sites are reduced to a reference point at 40°N, 141°E. The paleomagnetic data are divided into six time intervals, 33 Ma, 25–24 Ma, 23–20 Ma, 18–16 Ma, 14–10 Ma and 7–5 Ma (Table 2).

Table 2
Average paleomagnetic directions and paleomagnetic poles for NE Japan
Directions are presented as the values at a representative point in NE Japan (40°N, 141°E). All data are from welded tuffs. N = number of sites; D and I = mean declination and inclination respectively; k and α_{95} = Fisherian statistic parameters. R and ΔR = amount of rotation and its uncertainty with respect to East Asia. R is given as $D_{ob} - D_{ex}$, where D_{ex} is declination value expected from the apparent polar wander path for East Asia [40]. ΔR is given as $[(\Delta D_{ob})^2 + (\Delta D_{ex})^2]^{1/2}$, where ΔD_{ob} and ΔD_{ex} are the uncertainty at 95% confidence ($\Delta D = \sin^{-1}[\sin(\alpha_{95})/\cos I]$). A_{95} = radius of the circle of 95% confidence about the mean pole.

Age	N	D (°)	I (°)	α_{95} (°)	R (°)	ΔR (°)	Paleomagnetic pole Lat.	Long.	A_{95} (°)
33 Ma	3	-44.2	63.7	21.1	-42.3	54.3	57.6°N	78.4°E	26.3
27-25 Ma	8	-45.1	57.3	10.1	-43.2	19.0	56.3°N	65.7°E	13.5
23-20 Ma	11	-57.0	54.8	12.7	-58.3	22.5	47.5°N	67.1°E	15.7
18-16 Ma	7	-34.4	60.0	12.8	-35.7	26.4	62.7°N	67.6°E	16.8
14-10 Ma	7	-4.9	56.9	10.6	-6.2	19.8	85.4°N	43.2°E	12.0
7-5 Ma	6	2.0	58.9	11.4	0.8	22.4	88.3°N	173.3°W	14.1

Paleomagnetic directions prior to 20 Ma cluster around the westerly direction. The mean direction between 33 and 20 Ma is $D = -51.2°$, $I = 57.2°$ ($\alpha_{95} = 7.2°$). The northerly declination has appeared since 12 Ma, with the mean direction of data younger than 12 Ma being $D = 0.0°$, $I = 58.8°$ ($\alpha_{95} = 7.8°$). It is apparent that the declination change occurred between 20 and 12 Ma.

Tectonic rotation is estimated by comparison of observed declination (D_{ob}) and expected declination (D_{ex}), with the latter computed from the apparent polar wander path for East Asia [39,40]. The amount of rotation for a given sampling site is shown with its uncertainty as a function of age in Fig. 7. The amount of rotation (R) and its uncertainty (ΔR) are given as $D_{ob}-D_{ex}$ and $[(\Delta D_{ob})^2 + (\Delta D_{ex})^2]^{1/2}$, where ΔD_{ob} and ΔD_{ex} are the uncertainty at 95% confidence ($\Delta D = \sin^{-1}[\sin(\alpha_{95})/\cos I]$ with α_{95} the radius of the 95% confidence circle and I is the inclination). Data from two sites (NS515 and NS520) are discarded because their uncertainties exceed 20°. The first-order approximation for the rotation is described as (1) little rotation between 33 and 20 Ma (although fairly large variation in declination is observed in the interval 22–20 Ma) and (2) counterclockwise rotation of NE Japan between 20 and 12 Ma.

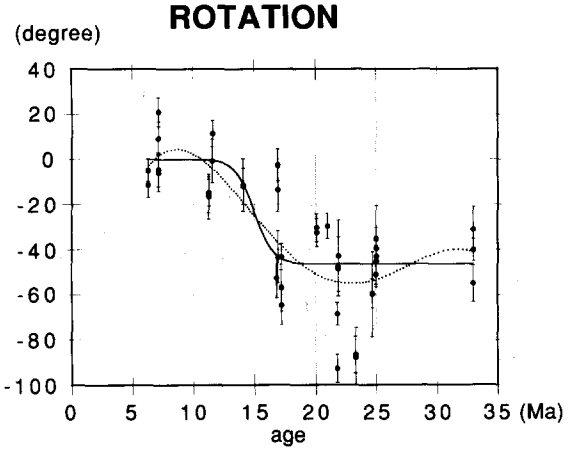


Fig. 7. Plot of rotation angle as a function of age. The error bars show uncertainty in rotation at the 95% confidence level (see text). The solid line shows the best-fit by a curve of the form $R(t) = 46.4[1-1/\{\exp((t-15)/0.75)+1\}]$, where $R(t)$ is the rotation angle. The time of the climax of the rotation is 15.0 Ma on the best-fit curve. The dotted line shows the the fourth-order polynomial fit.

Table 3
Parameters of the best-fit analysis
Rotation ($R(t)$) is fitted by the function $R(t) = R_o[1-1/\{\exp((t-t_o)/T)+1\}]$. Parameters t_o (Ma) and R_o (degrees) are determined by a non-linear least-squares fitting for given parameter T (m.y.). The parameters t_o and R_o are interpreted as the timing of the climax of the rotational motion and a net angle of the rotation respectively. Four times T ($4T$) is interpreted to be the duration of the rotation in which a block experiences 76% of its rotational motion.

T	to	Ro
0.01	14.1	44.4
0.1	14.2	44.4
0.5	14.6	44.7
0.75	15.0	46.4
1	15.1	48.2

Rotation versus age is fitted by a logistic curve for evaluating the rotational motion:

$$R(t) = R_o[1-1/\{\exp((t-t_o)/T) + 1\}]$$

where $R(t)$ is rotation change with respect to time (t), R_o corresponds to the net angle through which NE Japan was rotated, and t_o is the climax of the rotational motion [41]; T is a parameter ranging from 0.01 to 1 m.y. in this study. The best-fit function for the compiled data set is calculated by non-linear least-squares fitting. The parameters of the best-fit curve for a given T are listed in Table 3. One of the best-fit curves ($T =$ 0.75 m.y.) is drawn as a solid line in Fig. 7. The best estimate of the parameters R_o and t_o are −46.4° and 15.0 Ma for $T = 0.75$ m.y.

The most striking result from the best-fit curve for NE Japan is the timing of the climax of counterclockwise rotation, which is 15.0 Ma. This climax at 15 Ma is also suggested by the fourth-order polynomial fit dotted line in Fig. 7). Because the declination change occurred at a some time between 18–16 Ma and 10–14 Ma, the 15.0 Ma climax in the rotation is a reasonable estimate. This timing is greatly dependent on the westerly remanence declination of the Yashiozawagawa Formation (17.2 Ma), the Masukawa Formation (16.9 Ma) and the Ohtose Formation (16.8 Ma). The reliability of the westerly direction is

confirmed by the presence of normal polarities in the Ohtose Formation and reversed polarities in the Yashiozawagawa and Masukawa Formations. We therefore conclude that rotational motion of NE Japan reached a climax at about 15 Ma.

The timing of the climax of the counterclockwise rotation of NE Japan is compatible with the clockwise rotation of SW Japan. Paleomagnetism and K-Ar age determinations of Miocene volcanic rocks in SW Japan show that 80% of the rotational motion occurred between 16.1 and 14.2 Ma [4]. Paleomagnetism of the sedimentary rocks of the Setouchi Miocene series of central SW Japan suggests that nearly half of the rotation had been attained by the early Middle Miocene ($\sim$ 15 Ma) [42]. Curve-fitting analysis for a paleomagnetic dataset of dated volcanic and sedimentary rocks identifies the climax of the clockwise rotation of SW Japan at 14.9 Ma [41]. Within the limit of the age determinations the timing of rotation of NE Japan thus coincides with that of SW Japan.

Seeing as the recent results of ODP drilling in the Sea of Japan suggest that opening commenced at about 20 Ma [6,43–45], differential rotation of NE Japan and SW Japan is attributable to the opening of the Sea of Japan. The differential rotation of the Japan arc identified from paleomagnetism therefore constrains the timing of the opening of the Sea of Japan.

5.3. Amount of counterclockwise rotation

The best-fit curve identifies an amount of rotation of NE Japan of 46.4° in a counterclockwise direction. The mean calculation of the paleomagnetic directions between 33 and 20 Ma gives an alternative estimate of the amount of rotation. The mean declination is −51.2° at the reference point, whereas the expected declination at 27 Ma inferred from East Asia is −1.9° [40]. Hence the predicted rotation angle is 49.3° ($\Delta R = 13.4°$). These estimates imply that NE Japan was subjected to a large rotational motion of more than 45°.

The amount of rotation of the NE Japan arc is a clue to the opening mode of the northeastern part of the Sea of Japan. Limiting models are (A) a fan-shaped opening model [7] and (B) a parallel to subparallel opening model [1,5,46,47]. The former model requires rotational motion of the island arc, whereas the latter requires the island arc to move with little rotation. The paleomagnetic evidence from the large counterclockwise rotation of NE Japan strongly suggests that the fan-shaped opening is the most plausible mode for the northeastern part of the Sea of Japan. The fan-shaped opening mode is a possible mode for the back-arc spreading observed in several back-arc basins, such as the southwestern part of the Sea of Japan [48], the Valencia Trough, the Ligurian Sea in the western Mediterranean [49–52] and the Aegean Sea in the eastern Mediterranean [53].

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Appendix: Table A1
Magnetization directions of the low-temperature component. For notes, see Table 1

[Area] Formation site	(age) Demagnetization (°C)		N	in-situ D (°)	in-situ I (°)	tilt-corrected D (°)	tilt-corrected I (°)	k	α_{95} (°)	Rock Type	Locality Lat. (E)	Long. (N)
[Oshima Peninsula]												
Masukawa Formation												
HK7	B	0-200	8	-13.5	65.9	63.8	81.8	75.2	6.4	R	140°06.7'	42°36.0'
HK9	B	0-200	5	0.3	52.5	24.0	44.3	39.6	12.3	R	140°06.8'	42°35.4'
HK11	B	0-200	4	-2.9	58.7	27.6	57.6	94.7	9.5	R	140°06.8'	42°35.5'
Fukuyama Formation												
HK13	B	0-150	5	-6.0	61.9	34.0	33.3	132.5	6.7	R	140°13.5'	41°25.1'
HK15	B	0-200	11	6.2	55.8	39.3	16.4	39.0	7.4	R	140°13.0'	41°25.5'
HK17	C	0-400	5	7.2	64.5	43.1	33.7	102.5	7.6	R	140°12.5'	41°28.6'
[Tsugaru Peninsula]												
Gongenzaki Formation												
HK25	B	0-150	3	0.1	59.0	51.3	49.1	21.8	27.1	D	140°16.6'	41°08.1'
Ohtose Formation												
HK29	B	0-400	10	-12.6	65.8	-25.3	7.3	68.0	5.9	D	140°08.2'	40°38.7'
HK31	C	0-400	6	5.7	60.5	38.1	76.9	305.9	3.8	DT	140°07.4'	40°35.3'
[Akita]												
Yashiozawagawa Formation												
AK1	B	0-200	12	-5.6	59.3	38.4	51.3	145.4	3.6	R	140°12.6'	39°13.9'
AK2	B	0-150	5	-20.7	60.9	1.3	61.3	154.2	6.2	R	140°12.3'	39°13.5'
AK10	B	0-200	10	-7.6	59.2	43.2	52.3	93.6	5.0	R	140°12.4'	39°14.0'
[Ninohe]												
Nisatai Formation												
NS704	B	0-400	5	-37.6	58.5	-12.6	66.4	73.3	9.0	D	140°25.8'	38°19.1'

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